

App Features - Complete Requirements Document

User Roles

- Gym Owner
 - Participant
 - Trainer
 - Admin
 - Seller
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1. Chatbot Functionality

User Interaction

- Supports text, video, images, (voice optional)

Recommendation Generation

Must generate personalized workout plans, nutrition, goals, and suggestions using rules and ML models.

Feedback Learning

Should learn from user feedback, classify actions to improve future responses.

Form Correction (CV)

Motion Detection: The chatbot must capture the user's movements via device camera.

Posture Analysis: Use AI-based computer vision models to detect & evaluate exercise form.

(Optional) Real-time Feedback: Provide immediate on-screen feedback (e.g., suggestions to fix form).

(Optional) Real-time Text Feedback

User Privacy: Must request user's permission before enabling camera.

Conversational Features

NLP: Must support free-text queries like "Should I skip workout?"

Personalization Memory: Remember key user data (age, weight, workout) per session.

Multi-language Support: Spanish, French, and other languages always preferred.

Connect with Schedule Feature: When user sets workout plan, sync automatically to schedule table.

Additional Chatbot Capabilities

The chatbot will:

- Talk about various recipes and workouts.
- Provide fitness-related advice.
- Answer common boilerplate questions.
- Be restricted to fitness topics only (no unrelated conversations).

Actions with Direct Greetings

2. Schedule Feature

Workout Plan Table Example

Sun	Mon	Tue	Wed	Thu	Fri	Sat
Plan	Arms	Leg	Chest	Back	Free	Free
						Shoulders

Workout Alarm: 7:00 PM

Target Calories: 7000 cal

Features

- Notifications/alarms for workout time.
 - Reminder notifications for target calories and meals (user can set per day/time with specific calories).
 - Connected with chatbot so the user can ask the AI and make it insert/update the schedule (AI can insert/edit).
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3. e-Commerce Feature

Objectives

- Easy access to fitness products without leaving the app
- Increase user engagement and revenue through integrated commerce
- Deliver personalized experiences based on user behavior and fitness goals
- Ensure secure and user-friendly transactions
- Build trust through real pricing, reviews, and reliable fulfillment

Functional Requirements

A. Product Catalog & Browsing

Feature	Description
Product listing	Display all fitness products, including gym equipment, apparel/clothes, supplements, accessories, and wellness items.
Product details	Each product page includes: high-quality images, descriptions, pricing, shipping info, and options (size, flavor, etc.)
Search functionality	Users can search products by keyword, category, brand, or tags, with auto-suggest and filters.
Filtering & sorting	Filter results by price, range, rating, brand, or category. Sort by relevance, popularity, or newest.
In-app navigation	Add products to cart/wishlist without leaving current screen.
Product Reviews	Star rating, written reviews with timestamps, helpfulness voting (upvote/downvote).

B. Personalized Recommendations

Feature	Description
AI-based Recommendations	Suggest products in real-time using workout history, goals, behavior, purchases, preferences.
Contextual Suggestions	Show related items based on current browsing/activity context (homepage, chat, push, etc).
Frequently bought together	Display commonly bundled/purchased items (apply bundle discounts if possible).

C. Shopping Cart & Checkout

Step	Description
Add to Cart	Add, remove, or adjust quantities in cart. Cart persists for logged-in users.
Order Summary	Show summary before checkout: items, subtotal, shipping, total, discounts.
Payment Gateway	Secure payments (credit/debit cards, PayPal, Apple Pay, Google Pay).
PCI Compliance	Meets PCI-DSS standards to protect financial data.
Order Confirmation	Digital confirmation and receipt in-app and by email.

D. Shipping & Delivery

Feature	Description
Address Management	Add, edit, and save multiple shipping addresses.
Order Tracking	Provide tracking info/status within the app using courier APIs.
Notifications	Push/email notifications at each order stage (confirmed → dispatched → delivered).

E. Reviews & Rating

Feature	Description
Verified Reviews	Only buyers may submit reviews.
Content Moderation	AI/manual review must check for spam/inappropriate language before publishing reviews.
Rating Averages	Product pages show updated average rating and review count in real time.

4. Personalized Content Algorithm

The app's content feed should dynamically display posts, updates, and content that are most relevant to each individual user.

Relevance is determined via an AI-driven ranking algorithm that analyzes user behavior, preferences, and interactions.

This ensures every user's feed feels customized, fostering engagement and preventing irrelevant/repetitive content.

Functional Requirements

Requirement	Description
Personalized Ranking	Posts in the main feed must be ranked by a recommendation algorithm prioritizing relevance to user's interests and engagement patterns.
User Behavior Tracking	Continuously track engagement signals (likes, comments, shares, time spent viewing, categories interacted with, etc.).
Content Profiling	Tag each post with relevant metadata for accurate matching (hashtags, type, category, location, language).
Collaborative Filtering	Use collaborative filtering to recommend content liked by users with similar behavior/goals.
Contextual Prioritization	Consider contextual data (time of day, current goals, location) when ranking posts.
Feedback Loop	User actions update the model to refine and improve accuracy.
Diversity Control	Balance between personal preferences and new/trending content to avoid repetitive/bubble feeds.
Privacy Compliance	Data for personalization is anonymized and processed under explicit opt-in consent per GDPR or similar guidelines.

Algorithm Logic (Conceptual Explanation)

The feed uses a multi-factor weighted scoring system that evaluates the relevance of each post to a given user.

A simplified model might look like this:

Post Relevance Score = $(W_1 \times \text{Interest Match}) + (W_2 \times \text{Engagement History}) + (W_3 \times \text{Social Connections}) + (W_4 \times \text{Context Relevance}) + (W_5 \times \text{Popularity Weight})$

Where:

- **Interest Match:** Tags, hashtags, or topics matching the user's goals (e.g., yoga, bodybuilding).
- **Engagement History:** How often the user interacts with similar content types.
- **Social Connections:** Priority for posts from friends (optional if there's follow-back), trainers, or followed gyms.
- **Context Relevance:** Adjusted to the user's current routine, schedule, or time (e.g., "morning workout tips").
- **Popularity Weight:** Trending or high-performing posts across the community to maintain freshness.

A machine learning model can fine-tune these weights over time using historical data and behavior patterns (e.g., through reinforcement learning or ranking-based models).

5. Gym Owner Features

5.1 Manage Moderators

Purpose: Allow gym owners to delegate control to trusted moderators similar to Facebook page admins.

Details:

- Moderators can perform nearly all actions except user management (e.g., removing participants).
- No limit on the number of moderators.
- No approval or confirmation step is required when assigning/removing moderators.
- No activity logs or history tracking needed for moderator actions.

5.2 Manage Posts

Purpose: Allow gym owners (and moderators) to engage users through interactive content.

Details:

- Supports text, image, and video posts (like Facebook/LinkedIn).
- Posts can include tags/categories for better organization.
- Owners and moderators can both create, edit, and delete posts.
- Participants can react and comment on posts.
- All posts are published immediately (no scheduling).

5.3 Manage Trainers

- Add or Remove Trainers from the About Page

5.4 Manage Sessions

Purpose: Enable gyms to organize and promote classes or training programs.

Details:

- Session types include any gym-related activity (e.g., fitness, yoga, etc.).
- Each gym has a fixed number of rooms, and sessions must be assigned to a room.
- Sessions have limited participant capacity.
- Bookings are made via payment (not free).
- Both owners and trainers receive notifications or emails when a session is booked or canceled.

5.5 Monitor Reviews

Purpose: Maintain control and quality over gym feedback.

Details:

- Can reply publicly to reviews.
- Replies do not require approval or moderation.

5.6 Manage Gym Profile

Purpose: Allow gym owners to maintain up-to-date and engaging gym profiles.

Details:

- Owners can update all profile details, including name, bio, images, amenities, and pricing.
- Updates are applied instantly (no admin re-approval).
- Instead of social links, the app provides an in-app chat system for direct communication between participants and gyms.

5.7 Manage Bundles

Purpose: Allow gym owners to create flexible membership and subscription options.

Details:

- Supports monthly, annual, and custom bundles.
- Bundle description and details are written by the owner or moderator.
- Expired or unused bundles can be deleted permanently (no archival).

5.8 Manage Promo Codes

Purpose: Encourage customer engagement through discounts and special offers.

Details:

- Both owners and moderators can generate promo codes.
 - Each promo code includes discount %, expiration date, and usage limit.
 - Can apply to any gym-related item or bundle.
 - Redeemed only during checkout.
 - No need for promo code analytics or reporting.
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6. Participant Features

6.1 Browse Gyms

The system shall provide an interactive map view (powered by Google Maps or similar API) showing gyms around the user's current or selected location.

The user shall be able to switch between map and list view.

The system shall display live distance updates between the user and each gym if location access is granted.

Each gym marker shall show basic info (name, average rating, distance, price range).

When the user clicks View Profile, the system shall show full gym details including:

- Images (uploaded by gym owner only)
- Subscription prices
- Offers & promotions
- Ratings & reviews
- Amenities (pool, sauna, etc.)
- Gym type (male / female / mixed)
- Location and contact info
- "Last updated" timestamp

6.2 Search & Filtering

The participant shall be able to search gyms by name or location.

The participant shall be able to filter gyms by:

- Price range
- Distance
- Rating
- Gender type
- Amenities

The participant shall be able to sort results by:

- Price
- Distance
- Rating
- Subscription count

6.3 Gym Registration & Verification

- A user starts as a participant by default.
- To become a gym owner, a user must:
 - Submit ownership documents (e.g. business license, ID).
 - Await admin verification and approval.
- Only verified gym owners can create a gym profile and place their gym pin on the map.
- Gym information updates require re-approval before they become visible to users.

6.4 Offers & Promotions

Gym owners can create multiple offers at once (e.g. seasonal, referral, or bundled offers).

Offers must have:

- Description, discount value, start/end date.

Participants can apply promo codes at checkout when subscribing to a gym.

The system shall automatically expire offers and promo codes when they end or are used up.

Participants can receive notifications (email or in-app) for nearby offers.

As a participant, I want to see special offers and discounts available at gyms near me so that I can save money or try new places at lower cost.

I can:

- View all current offers (free trials, referral bonuses, seasonal discounts) directly on gym profiles.

- Receive push notifications for offers from gyms near my location or based on my saved preferences.
- See countdown timers for time-limited deals.

6.5 Rating and Reviews

Only authenticated participants can submit reviews.

Each review shall include:

- Star rating
- Text feedback (no images for now)

Reviews are always linked to the participant's profile (no anonymous feedback).

Participants can edit or delete their reviews.

Gym owners can reply to reviews publicly.

The system shall prioritize verified reviews (from users who booked or checked in).

A moderation system shall be implemented to flag and remove inappropriate content.

7. Trainer Features

7.1 Manage Trainers

Purpose: Allow verified trainers to create, manage, and promote their profiles and services.

Details:

- Trainers can register and await admin approval before their profile becomes visible.
- Profiles include name, bio, specialization, years of experience, certifications, and pricing.
- Trainers can edit or delete their profiles anytime (no re-approval needed after the first verification).
- Admins can deactivate or remove trainer profiles if needed.
- Trainers can set availability and time slots for their sessions.

7.2 Manage Sessions

Purpose: Enable trainers to create individual or group sessions that users can purchase and join.

Details:

- Trainers can create paid sessions with details: title, description, date, duration, and price.
- Sessions can be open (anyone can join) or group-based (limited participants).

- Trainers can specify deadlines for booking.
- Each session can include video materials and follow-up notes.
- Users can join sessions after successful payment.
- Trainers and users receive notifications upon booking or cancellation.
- Sessions cannot start without payment confirmation.

7.3 Manage Bundles

Purpose: Allow trainers to sell multiple related sessions together as a bundle.

Details:

- Bundles can include multiple sessions grouped under one theme or goal (e.g., "4-week Strength Program").
- Bundles must be paid immediately upon booking.
- Users cannot subscribe after the specified deadline.
- Trainers can edit or delete their bundles anytime.
- Each bundle has a title, description, duration, and price.
- Trainers can attach pre-recorded materials or guidance videos.
- Bundles are visible publicly to all participants once published.

7.4 Payment and Booking

Purpose: Handle secure payments and confirm bookings for sessions and bundles.

Details:

- Payments are processed immediately via integrated payment gateways.
- Booking is confirmed only after successful payment.
- Refunds are not supported in the MVP.
- Trainers receive a summary of their transactions.
- The system automatically deducts platform fees and credits the trainer's share.

7.5 Feedback and Ratings

Purpose: Enable participants to provide feedback and rate trainers or sessions to ensure transparency.

Details:

- Only participants who completed a session or bundle can leave feedback.
- Feedback includes both text and star ratings.

- Ratings are averaged and displayed on trainer profiles.
- Trainers cannot delete or edit reviews.
- Admins can remove reported or inappropriate reviews.

7.6 Participant Progress Tracking

Purpose: Allow trainers to track the progress and performance of their participants.

Details:

- Trainers can view participant lists per session or bundle.
 - Trainers can send follow-up messages and feedback in chat.
 - Progress tracking can be done via chat for MVP (no visual dashboard).
 - Participants can view trainer feedback from previous sessions.
 - Tracking can be canceled or ended by the trainer at any time.
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8. Communication Features

8.1 Chat System

Purpose: Enable seamless real-time communication between all user types (participants, trainers, and gym owners) to support engagement, inquiries, and collaboration.

Details:

- All account types can initiate and participate in chats (participants ↔ trainers ↔ gym owners).
- Supports 1-to-1 and group chats; any user can create a group chat.
- Messages support text and images by default; videos are optional (future enhancement).
- Features include:
 - Real-time message delivery
 - Read receipts and typing indicators
 - Persistent chat history (messages never expire unless deleted manually)
- Voice messages are not included in MVP.
- Future versions may support voice/video calls.
- End-to-end encryption required for privacy and GDPR compliance.
- Chat is text-only in the MVP (no media or file sharing).
- Messages are persistent and saved for each session or bundle.

- Trainers can start conversations with participants who purchased their sessions or bundles.
 - Chat logs are stored but not exportable.
 - Notifications are sent when new messages are received.
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9. Social Features

9.1 Posts and Feeds

Purpose: Provide a community-driven social feed for users to share fitness updates, progress, and motivational content.

Details:

- All account types (participants, trainers, and gym owners) can create posts.
- Posts support text and images (videos are excluded for MVP).
- Posts publish immediately without admin approval.
- Follow system: users can follow other users, trainers, or gyms to personalize their feed.
- Posts can include hashtags for categorization and discoverability (e.g., #FitnessGoals, #GymLife).
- Feeds are algorithmically curated based on user interests, interactions, and follows.
- Comments are enabled on posts.
- Sharing (internal or external) is not included in the MVP.
- Privacy settings are public-only for now (no private or friends-only posts).

9.2 Engagement Tools

Purpose: Encourage user interaction and retention through social features like reactions, comments, and notifications.

Details:

- Supports multiple reaction types (e.g., like, love, support).
- Users can follow any account type (participant, trainer, gym owner).
- In-app, push, and email notifications for:
 - New followers
 - Reactions
 - Comments
 - Mentions or replies

- Users can manage notification preferences (enable/disable certain alerts).
 - Notifications are persistent and stored in the user's account.
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10. Notifications

Purpose: Keep trainers and participants updated about booking, deadlines, and progress.

Details:

In-app and/or email notifications for:

- New booking confirmation
- Session start reminders
- Payment success
- Chat messages

Trainers receive alerts when a participant books or cancels a session.

Participants receive reminders before session deadlines.

11. Session and Bundle Discovery

Purpose: Allow participants to explore and join sessions or bundles easily.

Details:

- Participants can browse all available sessions and bundles by category or trainer.
 - Each listing displays: title, description, trainer name, rating, duration, and price.
 - Sorting and filtering options (by price, rating, or date) may be added later.
 - Sessions and bundles show countdown timers to indicate deadline.
 - Users can view trainer profiles directly from session pages.
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12. Access Control

Purpose: Define roles and permissions across different user types.

Details:

- Roles include: Admin, Trainer, and Participant.
- Only verified trainers can create sessions or bundles.

- Only logged-in participants can purchase or join sessions.
 - Admins can approve trainers, monitor sessions, and manage reports.
 - Trainers can only see data related to their own sessions.
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13. Admin Management

13.1 Admin Features

Purpose: Provide admins with full control over trainers, sessions, and reports.

Details:

- Admins can view all registered trainers and their certifications.
- Admins can approve, suspend, or delete trainers.
- Admins can manage all sessions and bundles on the platform.
- Admins can view system-level analytics (e.g., active trainers, total revenue).
- Admins can remove inappropriate reviews or reported content.

13.2 Trainer Verification

Purpose: Ensure only qualified and verified trainers can offer paid sessions.

Details:

- Trainers must upload certifications and credentials.
- Admins review and approve or reject trainer applications.
- Once approved, trainers gain access to session and bundle management.
- Re-verification may be required if trainers update critical info (e.g., certifications).